

AWS checklist and maintenance guide

Station name	QAS A	backbone
People present	JEB, MAV, pilot Simoa	instructed by:
Purpose	standard maintenance / installation / take-down / quick visit / emergency	
Date	23 Aug 2014	
Transport	helicopter / fixed wing / snow mobile / dog sleds / on foot from:	
Weather	warm / cold / very cold / no wind / some wind / rain / cloud cover %:	

Notes:

windy
3:40 PM

solar panel had been connected
in reverse /

Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC	+ / -
Adjust logger time to UTC (be sure PC runs on UTC time)	Y / N
Download data to PC and/or change CF Card (wait for green light)	Y / N
Name of logger program	
Photo or screen dump of values in fast scan mode	Y / N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	°
Mast tilt across boom looking from radiometer (- if clockwise)	°
Boom direction relative to north	° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.	Y / N
Radiometer aligned with mast and inclinometer? If not, measure.	Y / N
Temperature / humidity height (bottom of casing – surface)	cm
Sonic ranger on AWS height (membrane – surface)	cm
Wind sensor height (center of propeller – surface)	cm
Sonic ranger on stakes height (membrane – surface)	315 cm
Free length of stake – outer, holding boom with sonic ranger	337 cm
Free length of stake – middle	352 cm
Free length of stake – outer, holding boom without sonic ranger	326 cm
Free length of ablation hose on/above the ice surface	352 cm
Alt: depth of ablation hose using markings on hose	184, 189 cm
Height of ablation hose fluid level above the ice surface	cm
Vertical difference in surface height (at station – at hose) (approx)	cm
Length of thermistor string on surface from surface marking	cm
Height of surface irregularities	cm
Stake assembly still standing?	Y / N / more or less
Apparent damage to sensors? If yes, which?	Y / N

Notes:

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer			- Aligned with mast: Y/N - Clean: Y/N
Inclinometer			- Aligned with radiometer within 0.5°: Y/N
Satellite antenna			
Wind sensor			- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly			
T/hum assembly casing			- Fan spins and sounds OK: Y/N
T/hum assembly bracket			
Temperature probe in T/hum assembly			
HygroClip in T/hum assembly			- 40 °C offset in HygroClip temperature: Y/N
Sonic ranger on AWS			- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N
Solar panel			- Output OK: Y/N - Clean: Y/N
GPS antenna			- Stuck to top of logger enclosure: Y/N
Data logger			- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader			- LED green or flashing orange (active): Y/N
Multiplexer			- Clicking as it should: Y/N
Iridium modem IMEI number			- Transmissions tested: Y/N - Problems: Y/N
Barometer			
Sonic ranger on stakes			- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string			- Old string left in ice: Y/N/cut
Ablation hose assembly			- Old hose left in ice: Y/N/cut - Air in hose: Y/N/removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N		

Metadata after maintenance / establishment

Time logger clock set (to UTC)?	Y / N
Name of logger program	
Photo or screen dump of values in fast scan mode	Y / N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	°
Mast tilt across boom looking from radiometer (- if clockwise)	°
Boom direction relative to north (radiometer should point to true south)	° true/magn
Box on wind sensor exactly towards mast? If not, change.	Y / N
Radiometer aligned with mast and inclinometer? If not, align.	Y / N
Temperature / humidity height (bottom of casing – surface)	cm
Sonic ranger on AWS height (membrane – surface)	cm
Wind sensor height (center of propeller – surface)	cm
Sonic ranger on stakes height (membrane – surface)	cm
Free length of stake – outer, holding boom with sonic ranger	cm
Free length of stake – middle	cm
Free length of stake – outer, holding boom without sonic ranger	cm
Free length of ablation hose on/above the ice surface	cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	cm
Vertical difference in surface height (at station – at hose) (approx)	cm
Length of thermistor string on surface from surface marking	cm
Height of surface irregularities	cm

Notes: